

Project Proposal

Predicting Movie Success Based on Early Production Data

Project Overview

Use early movie production data (e.g., budget, cast, genre) to predict box office success.

Motivation

Movie production is costly. Predictive insights can help studios make better investment decisions.

Data Sources

- IMDb datasets
- Kaggle (The Movies Dataset)
- TMDB API

Methodology

- Data preprocessing
- Feature engineering
- Classification modeling (e.g., Logistic Regression, Decision Trees)

Tools & Technologies

Python, pandas, scikit-learn, Jupyter Notebook

Evaluation Metrics

Accuracy, F1 Score, ROC-AUC

Train/test split and cross-validation

Timeline

- Week 1-2: Data prep
- Week 3: EDA
- Week 4-5: Modeling
- Week 6: Reporting

Expected Outcome

A model that predicts box office success with good accuracy using early-stage data.